

do commitment Figure out where to show
42min "The Golem" - a tube - use from 23min on if not
time - last 20min

NUMBER FOUR: SENTIENT LIFE

1. INTRO AND CLIPS

This time we are starting a section on what it means to be a sentient being. How do I justify this in Religious terms? Simply that this issue is with us in ethics today in a big way. How we treat each other and the issues involved in genetic engineering, as well as the idea of building machines that think is involved.

SENTIENT comes from the latin, SENTIRE to percieve, or feel.

1. responsive or concious of sense impressions, 2. aware, or
3. finely sensitive in perception or feeling.

it has to do with feeling or sensation and most distinctly with moral powers, the ability to perceive moral issues and feeling or emotion in that context.

This takes in all beings, human or otherwise, or a able to discern conciously the reality around them and reflect on that.

In science fiction this theme is brought up often, especially in regard to created beings and the ethics of whether or not they should be created.

Show Turing Test - a tube

FRANKENSTEIN by Mary Shelly could be called the first science fiction novel as well as her first novel.

It was conceived in 1816 when her husband, percy Shelly, Lord Byron and herself agreed to write tales of the supernatural inspired by German ghost stories they had been reading.

Mary was the only one to complete her story and she told of how the idea had come from a nightmare she had.

She was not quite 19 when she began writing it and it was finished by 1818.

It is based on gothic tales of terror and on the medical practice called GALVANISM which used electric currents for medical treatment. Also included were the then known theories of the origins of life on Earth and the myth of Prometheus, who in greek myth brought fire to humans.

To the greek story is added the Roman myth of Prometheus plasticator who animates a human figure of clay.

Dr. Frankenstein creates a monster of terrible strength and bulk who inspires fear and loathing. It understands its rejection and when Frankenstein refuses to create a mate for it, it goes on a terrifying trail of murder.

Dr. Frankenstein pursues his creation to the arctic but dies trying to destroy it. The monster disappears into the frozen waste to commit suicide.

This is widely regarded as the origin of modern science fiction but is also a discourse on the concept of the "noble savage" in which essential goodness of natural being is corrupted by malevolence.

Later we will discuss another way of creating beings using magic of the Kaballah, this is the GOLOM, believed to have been created in Prague in 1599 by a Rabbi. If this account is right, this golom is lifelessly waiting re-animation in a synagoge attic in Prague. We will discuss this in HE, SHE AND IT.

OTHER BOOKS DEALING WITH SENTIENT HUMAN CREATIONS

I can't begin to mention all the works dealing with human creations that are sentient but here are a few.

FRANK HERBERT has a trilogy about a computer that thinks its God.

Beginning with DESTINATION; VOID and continuing with THE JESUS

INCIDENT AND THE LAZARUS EFFECT. Particularly in the JESUS

INCIDENT the ship which took humans to the stars now believes

humans should worship it. *Philip Dick's "Do Androids dream of electric sheep" (Blade Runner); and, of course, The New "Battlestar Galactica" w/ no Theretore Cylons and polytheistic humans*

CLIFFORD SIMAK has written several about robots who are sentient.

PROJECT POPE being the most prominent.

the end of a
Today we will watch ~~the~~ ST:NG episode which deals with the whole issue of what is sentient life. Data is involved in a hearing about whether he is a machine and thus property of Starfleet, or whether he is a sentient life form. ~~Because of the tight knit nature of this episode we will watch it in its entirety.~~

Melinda Snodgrass of Albuq. won a writers guild award for this episode which has been called a timeless tale of personal rights. *(show 5-8 (end) - @ 20 min - ~~Spa 2 on 2~~ Ep. 9-45 min.*

Following that, I will show a couple of clips concerning Data's attempt to create life and later, a discussion with his "mother" (Dr. Soong's wife) about his creation.

SHOW "MEASURE OF A MAN" discuss issues of slavery brought up by Guinen. also, idea of soul. - 20 min 5-8

skip- show clips from "the Offspring" (13 min), and Data's mother (+ min). discuss. Show "Short Circuit" clip - how this sci-fi movie determines if #5 is alive, ie, sentient.

see below the question in these clips is: Is it even right to create life, especially sentient life? Or are there too many things that can go wrong so we shouldn't even attempt it.

There are other trek episodes dealing with life created by humans and their rights:

QUALITY OF LIFE involves some smart computer tools called exocomps who Data determines have the ability to make decisions and show signs of sentient life. He defends their right to not be forced to do things which they know are dangerous.

EVOLUTION concerns 2 medical nanites who escape Wesley's experiment and get into the computer system where they multiply and evolve. Data is able to interface with them and determines their right to live. They are given a planet that is uninhabited to live and evolve upon.

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QUESTIONS TO DISCUSS

With so much that can happen in the attempt to create artificial intelligence, is it ethically right to even try?

"Measure of a man" brings up the issue of slavery. If we can create not one but many thinking machines, what should their rights be?

Are we liable to end up with machines like Herbert's SHIP or the computer HAL in 2001 who actually turn on their creators and rule over them?

can you think of any other books, shows, etc. dealing with this issue?

Terminator I, II, III
Battlestar Galactica
Blade Runner

SENTIENCE; PART II. POPE, VITANULS, HE SHE AND IT

These are three different types of stories dealing with sentience in different ways.

"Pope of the Chimps" deals with the evolution of a few of the great apes in to sentient beings. Questions involved here are involved with questions like: Does developing a religion make beings sentient?

"the Vitanuts" by John Brunner deals with the loss of sentience in humans. This is an eerie story and deals with the idea, found in many religious traditions, that there are only so many souls or spirits to go around. We will discuss that as we do the story.

Finally, He, She and It is concerned with the creation of sentient beings, one by magic, one by science. The issue of whether it is even ethical to create beings as tools will be discussed.

We will also watch a couple of clips, totaling about 18 minutes.

gone ~~TRON~~ a clip of about 13 minutes, shows what it means to enter into a computer, something like the people in He, She and It. it is interesting in that the movie was made in 1982, before ideas like virtual reality were a reality. Of course, the people tied into the net don't "physically" enter as Flynn does, but the idea of fully projecting oneself into the net is there.

show ~~SHORT CIRCUIT~~ I have a 4 minute clip of how the scientist determined how number 5 was alive. We will watch these following the discussion of our two short stories.

POPE OF THE CHIMPS - SUMMARY

Hal Vendelmans, one of the researchers in a project of working with Chimps and signing, finds out he is going to die of Leukemia.

He wants to use his death to teach the chimps that humans are not immortal.

Its decided that he can let the alpha chimps, that is, the most intelligent ones know he is dying. These include Leo, a 20 year old and the leader, and Grimsky, a 43 year old male.

Chimps begin meeting and using different signs that humans can't decipher. Some are recognized, "shirt", "hat", "human", "change", "banana" and "fly"

When Leo was asked directly what one of major new signs was he replied, "Jump high, come again"

researchers begin to think they may be evolving a religion.

One of the underlying purposes of the project was to understand how the 1st hominids crossed the intellectual boundry that seperates animals from humaniaty, ie, what makes one sentient.

ques: what is sientient? what is that boundry between animals and humanity? Does developing a religion make you human?

two things happen:

1. the chimps become more vocal, make more sounds.
2. Leo begins wearing Hal Vendelmans hat and shirt but only when the chimps are in a certain grove (nicknamed the Holy grove by the researchers).

humans have been called the "noisy animal"

Is it our vocalization that makes us human? The chimps seem to think so

How much of religion is sensitive rather than cognitive? Humans use incense, bells, bull roarers, special clothes. Is this necessary to religion?

Is it religion that makes us human, or is it because we are human that we have religion?

Does this make the chimps "human"?

the researchers ask the ques: Is Leo and the others playing at being human with Leo a high priest wearing the dead god's clothes giving him "divine attributes."?

Grimsky begins to die tho not ill.

there is a ritual distribution of meat while Leo wears Hal's hat. is this communion?

Leo gives Grimsky "last rites" by putting Hal's shirt over his (Grimsky's) shoulders.--uses new signs inter by humans as "ecclesiastical language"

Leo says: "Grimsky now be human being."

Soon other chimps, young ones, are found dead, what seems to be ritual murder.

When confronted, Leo says "When humans go away, they become a god, when chimps go away, they become humans."

Leo says god has given orders that all chimps are to become humans as soon as possible.

This pecking order is part of most life forms whether in the

barnyard, jungle, or among humans. It is humans who, being sentient, know it is part of life, most other species just live it out "the law of the jungle".

It is humans who project this pecking order to a life after death, that one moves to some higher level.

in the western traditions (Judaism, Christianity and Islam) there is a hope of resurrection to an immortal, imperishable body.

Momons believe they pre-exist and males will one day be gods ruling their own planets with their spirit wives.

those Christians who believe the end is coming hope never to die but simply be transformed while alive when Jesus comes again.

~~"Millennial fever" is heavy right now with the turn of the~~
millennium coming up.

if you are Buddhist or Hindu and/or believe in reincarnation you hope to have a better birth next life and eventually reach Mokse or Nirvana after thousands of lifetimes and thus never be born again but live in bliss always.

Thus, one would have to argue that the chimps developing a religion in which they believe they will become human, as humans become gods makes them fully sentient. Is this what sentients is all about? Or is there more to it.

Judy Veldaman, Hal's wife comes and speaks to Leo. tells him that Hal was looking down and Leo was sending to many chimps to god to soon. They are haveing to be put in a holding area she gives Leo Hal's wristwatch which Leo adds to his ecclessiastical paraphenalia.

there are no more killings but the chimps are still very much into their religion.

THE VITANULS
BY JOHN BRUNNER

This story is given from a Hindu point of view. It involves the ethics of overpopulation and extending life. This is an excellent story from this perspective looking at what would happen if more people lived on Earth than in all the previous time humans have been here.

Barry Chase, an American with the World Health Organization is being shown around a maternity hospital in India. He was supposed to be conducting a survey of the methods in use.

One doctor, a Dr. Kotiwala, is considered to be a virtual saint. This is his last day of delivering babies and he will then become a SUNNYASI, that is, a holy ascetic wanderer who leaves behind everything of this world.

The matron points out that nine months before this time there had been a big religious festival which people regarded as auspicious for increasing their families. Thus the hospital was packed with delivering mothers.

The matron explains that the doctor was influenced by the Jainas, a group still in existence who believe that AHIMSA or non-injury, to any living thing is wrong. The strictest Jainas wear cloth over their mouth and nose so as not to breathe in any living thing, or jiva as they called. They use brooms to carefully sweep any living things out of their path when walking.

They discuss the new drug which is an anti senility, anti death pill that is now on the market. Naturally everyone wants it.

In Hindu belief, the soul is reborn (called reincarnation) until it achieves perfection by eliminating all the bad karma and accumulating good. Then one is absorbed into the all, or BRACHMAN. Thus, death is not seen as an evil, but a stepping stone.

Dr. Kotiwala's last delivery he is spooked by the baby's eyes. Nothing can be found wrong, but he knows something is. When he looks in the baby's eyes he sees NOTHING.

About two years later Dr. Chance comes to find Kotiwala, now Ananda (which means bliss in sanskrit) Bhagat and catches up to him in a small village. Kotiwala has seen many children born with nothing there, including one in this village.

Since he was the first one to recognize what is now called vitanuls, or no life. What is meant is there is no mind, according to the western doctors but no soul would be closer.

There was an increasing sharp rise in these babies until about 80% were being born as congenital imbicils. This was true all over the world.

Everything that makes a person human, ie SENTIENT, had been left out. The anti senility treatment had been made public a few days after these doctors first meeting and everyone who could was getting in line for it.

To quote Dr. Chance:

Just as we were congratulating ourselves on sorting out the row about the anti-senility treatment, here comes the most fantastic crisis in history and it's not going to break, it's going to get worse, and worse, and worse. . .Over the past two weeks the rate has topped eighty percent. . .Out of every ten children born last week, no matter what country or continent, eight are MINDLESS ANIMALS."

As it turned out on checking the first kids with this imbicility were born on that particular day.

Dr. chance still doesn't get it. Kotiwala has to explain what happened. As many people are alive in this twenty-first century as have ever lived since the evolution of human beings. On the day they met, that number was exceeded for the first time.

He explains its not the lack of mind, but the lack of soul.

Chance still doesn't understand thinking it is absurd that there can be a finite number of souls stored in some cosmic warehouse.

Kotawala's answer is to go into the mountains and die so his soul is set free.

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DISCUSSION

This story calls medical ethics into question. There has been much discussion already about whether children everywhere should all be inoculated against disease. By doing this, there are more dying of famine and overpopulation is a serious problem.

It seems the more we try to beat mother nature at her game, the more calamities we bring upon ourselves from other places.

witness the fact that after only 60 years of using antibiotics they are already becoming ineffectual against many strains of bacteria.

Even if one doesn't believe in reincarnation as in this story, we are creating a hellish world for many people.

What is your reaction to this story?

In Kabbalah, especially Isaac Luria, and in Mormonism as well as others it is believed that all souls were created at the same time at the very beginning of time. There are a finite number.

What is your reaction to this idea?

a tube - "The Golem" - 42 min

HE, SHE AND IT by Marge Piercy - SYNOPSIS

This story takes place on two fronts. On the one hand it is placed about 60 years in the future in a world where the multinational corporations run things because most governments have fallen and in which a person gives themselves to a company for life much as they do in Japan today.

The other parallel story takes place in Prague, Czech. in 1599 and following. This is told by Shira's grandmother, Malkah, about her several greats back grandfather, ^{Rabbi Lowe} who was chief rabbi at that a time. It concerns how he made a GOLOM to protect the Jewish community there.

Shira works for one of these multis but is going thru a divorce. The company awards custody of her son, Ari, to the father. He takes Ari for a two year stint on a space station so Shira cannot even see him.

Shira decides to go home to her grandmother, who raised her and lives in what's known as a "free town". ie, towns that contract work for the corporations but are not connected to any of them. The Free towns have a "wrap" over them to protect from UV radiation. To go outside of it, one has to cover up or risk cancer.

The cities, now called GLOPS were partially under wraps, or domes, but the system had not been completed before the govt. stopped functioning. There were still elections, but politicians were ineffectual. Every quadrant was managed by the remains of the old UN, the eco-police.

There had been a great famine and plagues which killed 2 billion people and the eco police had authority over earth, water and air outside domes and wraps. otherwise, the multis ruled their enclaves and free towns defended themselves as they could. The glop rotted under the poisonous sky, ruled by feeding gangs and overlords. The ice caps were melting and many coastal areas were under water.

Everything was done thru the NET, computer hook ups which were built right into people's temples so one could simply plug in

to the computer net and do whatever they were trained to do. Computer literacy was an absolute must. Moving in and out of large Artificial Intelligence data bases was something taught to all children in the free towns but only the more educated in the glop. The ability to access the world's information and resculpt it was the equivalent of the difference between the propertied and the landless in the past .

This is the world of Shira and her grandmother (born in 1987).

Shira had been raised by her grandmother rather than her mother and, upon going home, learns why Her mother was not around during her childhood. She is an "information pirate" wanted by the multinationals for stealing information on drugs, etc., believeing knowledge should be free to all people.

Since Shira is a cybernetic expert, she goes to work with Avarrm, a man who has created a CYBORG, that is, a being made up of both biological and mechanical parts, who is programmed to protect their town. Yod, as he's called, looks human and is a secret experiment.

Their had been a world covenant that robots not resemble people as when they had there had been serious problems. People were afraid that machines would replace them, not in dangerous jobs but in well paid and comfortable jobs. Thus, mobile robots were to be obviously identifiable as machines and given only sufficient intelligence for their rote tasks. The penalty for building human-looking robots was blacklisting and death.

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Now we shift to Prague in 1599 where the cheif rabbi and kabbalist, Judah Loew, realizes they will need a protector.

There had been many problems for the Jews of Europe and with the arrival of the Dominican, Father Thaddeous, an inquisitor from Spain, he knows there will be more problems.

Rabbi Loew debates Thanddeus by invitation and it is a draw, which he knows is dangerous, especially with Good Friday and Easter coming.

After fasting and prayer, he has a dream in which a voice says "you must make a golom of clay to rise and walk the getto and save your people."

A GOLOM, is being in human form made not by ha-shem but by another human thru esoteric knowledge, particularly by the power of words and letters. The Sefer Yezirah, the mystical book of creation, contains what you must master to form a golem with the power of the Names of God and the power of letters and numbers. the word golom means "matter lump."

Kabbalistic tradition tells of many sages and saints who created a golem for a mystical rite. They would make and unmake these moving clay men, joining themselves to the power of creation, and in the chanting and the act, achieve ecstasy. It was one of the crowns of glory of a truly holy person.

Sages also were said to make golems for some private use like running messages, cleaning the house, etc. like robots. These were both mute and stupid.

Rabbi Lowe hesitated but when the silversmith was arrested on a trumped up charge, he knew he had to proceed. He took two men with him, snuck out of the getto to the graveyard and they began digging clay while Loew chanted in a low sing song.

When they had enough clay he lay down a wooden form shaped like a man and packed clay over the shape. the shape was as tall and broad as the largest soldier.

He begins to pray, rocking and chanting. He feels the power of creation coming thru him. He chants all the combinations of letters and vowels and the hidden names of God, the sacred numbers that built the atoms of the universe.

Blue fire crackles over the clay doll as if rivulets of mercury ran over the surface and then sank inside. The clay begins to smoke and to heat dull red and then brighter and brighter. It turns into flesh before their eyes.

they chant and carry the Torah around the golom seven times and the golom begins to breath, open his eyes. Lowe puts the mark of ha shem on his forehead which gives him life.

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The rabbi names him Joseph and begins learning by first learning to walk so they can get back into the ghetto before dawn.

The golom accidently kills 2 watchmen but as the golom says, he is there to protect.

Rabbi Lowe begins teaching Joseph what he is to do to act like a man. Like eating, for him a brick and a potato are the same. He begins to go out at night to patrol and spy.

2060 Shira begins teaching Yod how to act more human since his predecessors went berserk before they could be trained.

Malkah tells Shira that Yod is a person, not human, but a person none the less. Malkah had helped program him.

She begins training Yod and finds he can feel pain and pleasure. She asks, "Do you consider yourself alive?"

He says, "I'm conscious of my existence, I think, I plan, I feel, I react. I consume nutrients and extract energy from them. I grow mentally, if not physically. I feel the desire for companionship. . . ."

Yod knows that Avarram had destroyed all the cyborgs before him. They were all conscious, fully alive minds.

He is upset saying: "If your mother had killed eight siblings of yours before your birth because they didn't measure up to her ideas of what she wanted, wouldn't you be alarmed.?" He fears being destroyed also.

Yod begins doing what he was meant to do, be hooked into the Net to moniter. Five people had been assassinated and made into vegetables due to attacks.

embedded in the base, plugged in, a person was vulnerable to mental warfare. The mind could be forced into a catotonic loop. A program could be launched that stoped breath or the brain shocked to death. Traps could be set by pirates or raiders which could ambush and kill and steal creations. Raiders came from both the multtis and free lance pirates.

by monitering the base, Yod saved Malkah from this fate. He remains plugged into the net for days. Malkah is afraid to get back on the net. Yod succeeded in killing both of the raiders.

people from the multi that Shira had belonged to. She knew them to be assassins for her cooperation.

Since they have been accessed, they must reconstruct the base and put in other security. Meanwhile, Yod and Shira begin becoming intimate, even though he is a cyborg.

Riva, Shira's mother arrives with her lover, Nili who is a Palestinian. Nili is also made up of a lot of machine parts, but is essentially human, albeit, an enhanced human.

Nili points out that the ability to read and write belonged to the church except for heretics and Jews. The Jews, as people of the book, have always considered getting knowledge part of being human. However, in their own day, few read, most plug into virtual reality stimulies to have experience and few plug in and access information on a scale never before available. The many know less and less and the few more and more.

Y-S, Shira's multi, asks her to meet with them and come back to work, saying she can have Ari if she does. However, it is a set up and Riva is "killed" and Yod hurt. Shira is afraid they know about Yod. However, there is no body for Riva. The fact is, this was staged so she would be thought dead. However, Malkah and Shira won't know this for some time.

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PRAGUE, 1600

Joseph has discovered on his nightly outings that there is a plan to invade the Ghetto in the weeks for Good Friday. The Rabbi decides to appeal to Johannes Kepler and other scientists for help. They find one of their group has been burned for heresy in Rome recently. However, they get a letter which will allow them into see the prince, and hopefully get help.

Joseph Golom goes, takes a jewel to the prince, and begs for help. However, all he gets is "I'll see what I can do" from the prince. On the way home he is accosted by robbers and ends up killing them all, including a woman. The Rabbi is not happy. However, in the process he learns about a weapon, a club, he can make to use for protection.

"A people in trouble is perceived as a troublesome people." Thus word comes that the emporer is doing what he can to stop trouble but he can't prevent Fr. Thaddeus from leading a good friday procession to the gates of the Ghetto. The emporer waffles since he can't have soldiers fire on a religious procession.

Joseph and the men try to shore up their defenses as best they can. He organizes as many as will fight, tho some will not do violence. As Joseph watches the rabbi he thinks, like father, like son, knowing that Rabbi Lowe would not appreciate that.

As Joseph thinks about the battle he realizes he doesn't know if he can die. He knows he can be wounded, but he heals fast.

The few soldiers try to stop the mob, but the huge number wins out and a battering ram is brought and after a time, batters down the gate. Joseph grabs it and uses it as a club. They lose some fifty dead and many wounded but the battle of 'Good Friday' is over, thanks to Joseph.

Joseph is a hero of the ghetto but The Rabbi realizes that the time of danger is passing, since the emporer has gotten rid of Thaddeus and a more accomidating dominican is promised. He worries that the longer Joseph remains in the world, the more he will come to regret it, and age is catching up with him.

the decision is made to "unmake" Joseph. They have to prepare with prayer and fasting. Joseph, upon trying to leave on his own suddenly feels weak. He feels downcast and afraid of losing his strength.

The Rabbi and other two men tell Joseph they have a task in the attic to perform. Joseph is puzzled but Rabbi Lowe is sure he must undo what he has done. Joseph obeys by lying down, but says "I have't done anything bad, I carried out what you wanted, I want to live, I want to be a man."

However, The rabbi is chanting and circling him with the Torah and Joseph can't rise. He fights dying saying, No, No. I deserve to live. I fought for you, saved you, my life is sweet to me.

The rabbi chants all the gates of the holy name and alphabet running backwards now. Slowly Joseph stops speaking, his features blur, he looks like he is melting. His face looks like a rock and his fingers become one lump, but there the process stops.

The Rabbi fabricates a story, covers him with a sheet, and padlock the door to the attic so noone can come in. He can be brought back to life again if the need arises.

Shortly thereafter, Rabbi Lowe dies. AS Malkah relates, tho the need for him has arisen, the knowledge of how to awaken him is lost. till now (this refering to Yod).

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2060

Malkah mourned Riva's death. Word comes from her multi that her husband and son are now back in Nebraska in the enclave and apologies for the "assasins" who killed her mother and others.

They decide to send a message robot to Ari, for his birthday. When it return they here disturbing voices in the background telling Ari to make his mom come.

Malkah, Avarrm and Shira decide they must enter the Y-S base thru their computers, using Yod. They need the information on its war plans and aims.

They enter the Y-S enclave and thru virtual reality it is as if they are really there, flying over the area. Yod teaches them how to become transparent so they won't be seen, but they are attacked by metallic harpies, and this is as real as if it were physical. Yod also teaches them to turn themselves into whatever they need, digging machines, or whatever.

Yod finds and stores the information they need. then they left a computer virus. They make it out, exausted but successful. On examining the material they find that Avarrm and his work are the cheif target. They question Shiras ties to her grandmother and, of course, her mothers activiites.

Shira realizes that until she gets her son back, she is vulnerable. A trip to the glop, where Nili has contacts and

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possible allies might be found for Shira, et al is planned. They bargain and fight their way to a group in the glop Nili knows and it is here Shira discovers her mother, not dead at all. It was, in fact because of her mother that they had not been killed while trying to find this group.

After establishing allies in the glop for war with Y-S, Yod and Shira head for the Y-S compound in Nebraska to kidnap Ari. It is a difficult operation, but they succeed, even tho Josh, Shira's X, is killed in the process. They return to Tikva, her town, with Ari and realize the war is not over with Y-S.

a complaint had been filed that Yod was not paid. The city council was going to find out what he really was. They would have to vote on whether he was a citizen of Tikva or Avarrms tool.

He says at the meeting: "I'm a cyborg but I am also a person. I think and feel and have existence just as you do."

The decision had to be made, and his nature known within the week as an attack from Y-S was expected soon.

Y-S wants to extradite Yod as a murderer. However, they have reason to believe he is a cyborg and want him for that reason. They meet with Y-S thru the net in a conference room. Josh, her X is there and she realizes he's not dead.

They simply say if Avarrm, et. al. don't turn over the cyborg they will send assassins every week into their base so all their effort will be used in defending and rebuilding. It would ruin their economy.

They want Yod to become the progenitor of a race.

Shira asks Josh a question which he answers wrongly, and she realizes its not him. They want Shira back to handle the new cyborgs they will build but she sees this as traficing in people.

The city council must decide if Yod is property or a citizen with the free will to decide if he wants to go to Y-S. Yod writes a speech to deliver to the council the next night. The council split down the middle so no decision was made.

Avarrm programs Yod to self destruct and take as many Y-S people with him as possible. It is his choice when he can do it. Yod makes sure Shira and Malkah are home, not in Avarams lab while he is taken and during his self destruct sequence.

Yod agrees to this as he realizes he is protecting the town and the people he cares for by doing so and this is important to him. He goes with the Y-S security people in the morning.

Nili comes in while they wait for word of Yod. Malkah says to her: "Yod was a mistake. You're the right path, Nili. It's better to make people into partial machines than to create machines that feel and yet are still controlled like cleaning robots. The creation of a conscious being as any kind of tool--supposed to exist only to fill our needs--is a disaster."

Suddenly there is a huge explosion, they run and discover that Avarrm's lab has exploded and he has been killed. All his notes and work of the past 20 years were gone.

Shira access the message Yod had told her to after he died. He tells her that he has taken Avarrm and his lab and records of his experiments with him in death. "I want there to be no more weapons like me. A weapon should not be conscious. A weapon should not have the capacity to suffer for what it does, to regret, to feel guilty."

He says he wants to be sure there are no more conscious beings created. That is the one good thing he feels he has done with his death.

Later Malkah reflects on it saying: "It was inexcusable to create a sentient being for any other reason than to live its own life. In the myth of Pygmalion, we assume that she would love her sculptor, but Shaw knew better. Each one of us wants to possess ourselves; only fools willingly give themselves away. Slavery produces the slave revolt."

Shira discovers Avramms logs at Malkah's home, briefly toy with the idea of recreating Yod as a companion. She realizes that is wrong and carries the crystals etc. to the recycling plant. She had set him free.

HE, SHE AND IT - CONCLUSIONS

Like many of the other stories, this one deals with the ethical delima of creating life to sereve us. The fascinating thing about this one is the magical creation of a GOLOM which is part of the kabbalistic lore. It is believed that Rabbi Lowe did indeed create a golom who is still in a synagog attic in Prague.

What are your reactions to this story?

Is it ethical to "play God" in creating life?

What about the future that Piercy sees, is it a likely one?

* * * * *

CONCLUSIONS TO WHOLE SECTION

We have discussed a number of issues in the whole.

1. we tend to see sentience in terms of what we humans see as traits that are like us.
2. there is also a tendency to superimpose human morality, whatever system we come from, upon our understanding of sentience in others.
3. Is it right, or ethical, to create life in our own image. If the writer sees humanity as less than perfect, does the created being show the same imperfection? Frankenstein is a good case and point.
4. Does creation determine the right to control or destroy what has been created? Are they property or free to choose?
5. How much does ones religious presuppositions determine how the story will evolve?

The Pope of the Chimps

Robert Silverberg

*For a quarter of a century, Robert Silverberg's stories and novels have set a standard for other writers of speculative fiction to match, a level of achievement attested by numerous awards and by both popular and critical acclaim. He tells, in clean and graceful prose, stories that are exciting as much for their ideas as for their action. His work ranges from the moving psychological novel, *Dying Inside*, to his recent epic fantasy, *Lord Valentine's Castle*.*

*Silverberg has frequently explored religious ideas in novels such as *Downward to the Earth*, *Tower of Glass*, and *The Book of Skulls*, and such widely read stories as "Good News from the Vatican," "The Feast of St. Dionysus," and "Born With the Dead." In "The Pope of the Chimps," he uses a contemporary topic and setting to raise some universal and thought-provoking questions.*

Early last month Vendelmans and I were alone with the chimps in the compound when suddenly he said, "I'm going to faint." It was a sizzling May morning, but Vendelmans had never shown any sign of noticing unusual heat, let alone suffering from it. I was busy talking to Leo and Mimsy and Mimsy's daughter Muffin and I registered Vendelmans's remark without doing anything about it. When you're intensely into talking by sign language, as we are in the project, you sometimes tend not to pay a lot of attention to spoken words.

But then Leo began to sign the trouble sign at me and I turned around and saw Vendelmans down on his knees in the grass, white-faced, gasping, covered with sweat. A few of the chimpanzees who aren't as sensitive to humans as Leo is thought it was a game and began to pantomime him, knuckles to the ground and bodies going limp. "Sick—" Vendelmans said. "Feel—terrible—"

I called for help and Gonzo took his left arm and Kong took his right and somehow, big as he was, we managed to get him out of the compound and up the hill to headquarters. By then he was complaining about sharp pains in his back and under his arms, and I realized that it wasn't just heat prostration. Within a week the diagnosis was in.

Leukemia.

They put him on chemotherapy and hormones and after ten days he was back with the project, looking cocky. "They've stabilized it," he told everyone. "It's in remission and I might have ten or twenty years left, or even more. I'm going to carry on with my work."

But he was gaunt and pale, with a tremor in his hands, and it was a frightful thing to have him among us. He might have been fooling himself, though I doubted it, but he wasn't fooling any of us: to us he was a *memento mori*, a walking death's-head-and-crossbones. That laymen think scientists are any more casual about such things than anyone else is something I blame Hollywood for. It is not easy to go about your daily work with a dying man at your side—or a dying man's wife, for Judy Vendelmans showed in her frightened eyes all the grief that Hal Vendelmans himself was repressing. She was going to lose a beloved husband unexpectedly soon and she hadn't had time to adjust to it, and her pain was impossible to ig-

nore. Besides, the nature of Vendelmans's dyingness was particularly unsettling, because he had been so big and robust and outgoing, a true Rabelaisian figure, and somehow between one moment and the next he was transformed into a wraith. "The finger of God," Dave Yost said. "A quick flick of Zeus's pinkie and Hal shrivels like cellophane in a fireplace." Vendelmans was not yet forty.

The chimps suspected something too.

Some of them, such as Leo and Ramona, are fifth-generation signers, bred for alpha intelligence, and they pick up subtleties and nuances very well. "Almost human," visitors like to say of them. We dislike that tag, because the important thing about chimpanzees is that they *aren't* human, that they are an alien intelligent species; but yet I know what people mean. The brightest of the chimps saw right away that something was amiss with Vendelmans, and started making odd remarks. "Big one rotten banana," said Ramona to Minisy while I was nearby. "He getting empty," Leo said to me as Vendelmans stumbled past us. Chimp metaphors never cease to amaze me. And Gonzo asked him outright: "You go away soon?"

"Go away" is not the chimp euphemism for death. So far as our animals know, no human being has ever died. Chimps die. Human beings "go away." We have kept things on that basis from the beginning, not intentionally at first, but such arrangements have a way of institutionalizing themselves. The first member of the group to die was Roger Nixon, in an automobile accident in the early years of the project, long before my time here, and apparently no one wanted to confuse or disturb the animals by explaining what had happened to him, so no explanations were offered. My second or third year here Tim Lippinger was killed in a ski-lift failure, and again it seemed easier not to go into details with them. And by the time of Will Bechstein's death in that helicopter crackup four years ago the policy was explicit: we chose not to regard his disappearance from the group as death, but mere "going away," as if he had only retired. The chimps do understand death, of course. They may even equate it with "going away," as Gonzo's question suggests. But if they do, they surely see human death as something quite different from chimpanzee death—a translation to another state of being, an ascent on a chariot of fire. Yost believes that they have no comprehension of human death at all, that they think we are immortal, that they think we are gods.

* * *

Vendelmans now no longer pretends that he isn't dying. The leukemia is plainly acute and he deteriorates physically from day to day. His original this-isn't-actually-happening attitude has been replaced by a kind of sullen angry acceptance. It is only the fourth week since the onset of the ailment and soon he'll have to enter the hospital.

And he wants to tell the chimps that he's going to die.

"They don't know that human beings can die," Yost said.

"Then it's time they found out," Vendelmans snapped. "Why perpetuate a load of mythological bullshit about us? Why let them think we're gods? Tell them outright that I'm going to die, the way old Egbert died and Salami and Mortimer."

"But they all died naturally," Jan Morton said.

"And I'm not dying naturally?"

She became terribly flustered. "Of old age, I mean. Their life-cycles clearly and understandably came to an end, and they died, and the chimps understood it. Whereas you—" She faltered.

"—am dying a monstrous and terrible death midway through my life," Vendelmans said, and started to break down, and recovered with a fierce effort, and Jan began to cry, and it was generally a bad scene, from which Vendelmans saved us by going on, "It should be of philosophical importance to the project to discover how the chimps react to a revaluation of the human metaphysic. We've ducked every chance we've had to help them understand the nature of mortality. Now I propose we use me to teach them that humans are subject to the same laws they are. That we are not gods."

"And that gods exist," said Yost, "who are capricious and unfathomable, and to whom we ourselves are as less than chimps."

Vendelmans shrugged. "They don't need to hear all that now. But it's time they understood what we are. Or rather, it's time that we learned how much they already understand. Use my death as a way of finding out. It's the first time they've been in the presence of a human who's actually in the process of dying. The other times one of us has died, it's always been in some sort of accident."

Burt Christensen said, "Hal, have you already told them anything about—"

"No," Vendelmans said. "Of course not. Not a word. But I see them talking to each other. They know."

* * *

We discussed it far into the night. The question needed careful examination because of the far-reaching consequences of any change we might make in the metaphysical givens of our animals. These chimps have lived in a closed environment here for decades, and the culture they have evolved is a product of what we have chosen to teach them, compounded by their own innate chimpness plus whatever we have unknowingly transmitted to them about ourselves or them. Any radical conceptual material we offer them must be weighed thoughtfully, because its effects will be irreversible, and those who succeed us in this community will be unforgiving if we do anything stupidly premature. If the plan is to observe a community of intelligent primates over a period of many human generations, studying the changes in their intellectual capacity as their linguistic skills increase, then we must at all times take care to let them find things out for themselves, rather than skewing our data by giving the chimps more than their current concept-processing abilities may be able to handle.

On the other hand, Vendelmans was dying right now, allowing us a dramatic opportunity to convey the concept of human mortality. We had at best a week or two to make use of that opportunity; then it might be years before the next chance.

"What are you worried about?" Vendelmans demanded.

Yost said, "Do you fear dying, Hal?"

"Dying makes me angry. I don't fear it; but I still have things to do, and I won't be able to do them. Why do you ask?"

"Because so far as we know the chimps see death—chimp death—as simply part of the great cycle of events, like the darkness that comes after the daylight. But human death is going to come as a revelation to them, a shock. And if they pick up from you any sense of fear or even anger over your dying, who knows what impact that will have on their way of thought?"

"Exactly. *Who knows?* I offer you a chance to find out!"

By a narrow margin, finally, we voted to let Hal Vendelmans share his death with the chimpanzees. Nearly all of us had reservations about that. But plainly Vendelmans was determined to have a useful death, a meaningful death; the only way he could face his fate at all was by contributing it like this to the project. And in the end I think most of us cast our votes his way purely out of our love for him.

We rearranged the schedules to give Vendelmans more contact with the animals. There are ten of us, fifty of them; each of us has a special field of inquiry—number theory, syntactical innovation, metaphysical exploration, semiotics, tool use, and so on—and we work with chimps of our own choice, subject, naturally, to the shifting patterns of subtribal bonding within the chimp community. But we agreed that Vendelmans would have to offer his revelations to the alpha intelligences—Leo, Ramona, Grimsky, Alice, and Attila—regardless of the current structure of the chimp/human dialogues. Leo, for instance, was involved in an on-going interchange with Beth Rankin on the notion of the change of seasons. Beth more or less willingly gave up her time with Leo to Vendelmans, for Leo was essential in this. We learned long ago that anything important had to be imparted to the alphas first, and they will impart it to the others. A bright chimp knows more about teaching things to his duller cousins than the brightest human being.

The next morning Hal and Judy Vendelmans took Leo, Ramona, and Attila aside and held a long conversation with them. I was busy in a different part of the compound with Gonzo, Mimsy, Muffin, and Chump, but I glanced over occasionally to see what was going on. Hal looked radiant—like Moses just down from the mountain after talking with God. Judy was trying to look radiant too, working at it, but her grief kept breaking through: once I saw her turn away from the chimps and press her knuckles to her teeth to hold it back.

Afterward Leo and Grimsky had a conference out by the oak grove. Yost and Charley Damiano watched it with binoculars, but they couldn't make much sense out of it. The chimps, when they sign to each other, use modified gestures much less precise than the ones they use with us; whether this marks the evolution of a special chimp-to-chimp argot designed not to be understood by us, or is simply a factor of chimp reliance on supplementary non-verbal ways of communicating, is something we still don't know, but the fact remains that we have trouble comprehending the sign language they use with each other, particularly the form the alphas use. Then, too, Leo and Grimsky kept wandering in and out of the trees, as if perhaps they knew we were watching them and didn't want us to eavesdrop. A little later in the day Ramona and Alice had the same sort of meeting. Now all five of our alphas must have been in on the revelation.

Somehow the news began to filter down to the rest of them.

We weren't able to observe actual concept transmission. We did notice that Vendelmans, the next day, began to get rather more attention than normal. Little troops of chimpanzees formed about him as he moved—slowly, and with obvious difficulty—about the compound. Gonzo and Chump, who had been bickering for months, suddenly were standing side by side staring intently at Vendelmans. Chicory, normally shy, went out of her way to engage him in a conversation—about the ripeness of the apples on the tree, Vendelmans reported. Anna Livia's young twins Shem and Shaun climbed up and sat on Vendelmans's shoulders.

"They want to find out what a dying god is really like," Yost said quietly.

"But look there," Jan Morton said.

Judy Vendelmans had an entourage too: Minsy, Muffin, Claudius, Buster, and Kong. Staring in fascination, eyes wide, lips extended, some of them blowing little bubbles of saliva.

"Do they think she's dying too?" Beth wondered.

Yost shook his head. "Probably not. They can see there's nothing physically wrong with her. But they're picking up the sorrow-vibes, the death-vibes."

"Is there any reason to think they're aware that Hal is Judy's mate?" Christensen asked.

"It doesn't matter," Yost said. "They can see that she's upset. That interests them, even if they have no way of knowing why Judy would be more upset than any of the rest of us."

"More mysteries out yonder," I said, pointing into the meadow.

Grimsky was standing by himself out there, contemplating something. He is the oldest of the chimps, gray-haired, going bald, a deep thinker. He has been here almost from the beginning, more than thirty years, and very little has escaped his attention in that time.

Far off to the left, in the shade of the big beech tree, Leo stood similarly in solitary meditation. He is twenty, the alpha male of the community, the strongest and by far the most intelligent. It was eerie to see the two of them in their individual zones of isolation, like distant sentinels, like Easter Island statues, lost in private reveries.

"Philosophers," Yost murmured.

Yesterday Vendelmans returned to the hospital for good. Before he went, he made his farewells to each of the fifty chimpanzees, even

the infants. In the past week he has altered markedly: he is only a shadow of himself, feeble, wasted. Judy says he'll live only another few weeks.

She has gone on leave and probably won't come back until after Hal's death. I wonder what the chimps will make of her "going away," and of her eventual return.

She said that Leo had asked her if she was dying too.

Perhaps things will get back to normal here now.

Christensen asked me this morning, "Have you noticed the way they seem to drag the notion of death into whatever conversation you're having with them these days?"

I nodded. "Mimsy asked me the other day if the moon dies when the sun comes up and the sun dies when the moon is out. It seemed like such a standard primitive metaphor that I didn't pick up on it at first. But Mimsy's too young for using metaphor that easily and she isn't particularly clever. The older ones must be talking about dying a lot, and it's filtering down."

"Chicory was doing subtraction with me," Christensen said. "She signed, '*You take five, two die, you have three.*' Later she turned it into a verb: '*Three die one equals two.*'"

Others reported similar things. Yet none of the animals were talking about Vendelmans and what was about to happen to him, nor were they asking any overt questions about death or dying. So far as we were able to perceive, they had displaced the whole thing into metaphorical diversions. That in itself indicated a powerful obsession. Like most obsessives, they were trying to hide the thing that most concerned them, and they probably thought they were doing a good job of it. It isn't their fault that we're able to guess what's going on in their minds. They are, after all—and we sometimes have to keep reminding ourselves of this—only chimpanzees.

They are holding meetings on the far side of the oak grove, where the little stream runs. Leo and Grimsy seem to do most of the talking, and the others gather around and sit very quietly as the speeches are made. The groups run from ten to thirty chimps at a time. We are unable to discover what they're discussing, though of course we have an idea. Whenever one of us approaches such a gathering, the chimps very casually drift off into three or four separate groups and look exceedingly innocent—"We just out for some fresh air, boss."

Charley Damiano wants to plant a bug in the grove. But how do you spy on a group that converses only in sign language? Cameras aren't as easily hidden as microphones.

We do our best with binoculars. But what little we've been able to observe has been mystifying. The chimp-to-chimp signs they use at these meetings are even more oblique and confusing than the ones we had seen earlier. It's as if they're holding their meetings in pig-latin, or doubletalk, or in some entirely new and private language.

Two technicians will come tomorrow to help us mount cameras in the grove.

Hal Vendelmans died last night. According to Judy, who phoned Dave Yost, it was very peaceful right at the end, an easy release. Yost and I broke the news to the alpha chimps just after breakfast. No euphemisms, just the straight news. Ramona made a few hooting sounds and looked as if she might cry, but she was the only one who seemed emotionally upset. Leo gave me a long deep look of what was almost certainly compassion, and then he hugged me very hard. Grimsky wandered away and seemed to be signing to himself in the new system. Now a meeting seems to be assembling in the oak grove, the first one in more than a week.

The cameras are in place. Even if we can't decipher the new signs, we can at least tape them and subject them to computer analysis until we begin to understand.

Now we've watched the first tapes of a grove meeting, but I can't say we know a lot more than we did before.

For one thing, they disabled two of the cameras right at the outset. Attila spotted them and sent Gonzo and Claudius up into the trees to yank them out. I suppose the remaining cameras went unnoticed; but by accident or deliberate diabolical craftiness, the chimps positioned themselves in such a way that none of the cameras had a clear angle. We did record a few statements from Leo and some give-and-take between Alice and Anna Livia. They spoke in a mixture of standard signs and the new ones, but, without a sense of the context, we've found it impossible to generate any sequence of meanings. Stray signs such as "shirt," "hat," "human," "change," and "banana fly," interspersed with undecipherable stuff, *seem* to be adding up to something, but no one is sure what. We observed no mention of Hal

Vendelmans nor any direct references to death. We may be misleading ourselves entirely about the significance of all this.

Or perhaps not. We codified some of the new signs and this afternoon I asked Ramona what one of them meant. She fidgeted and hooted and looked uncomfortable—and not simply because I was asking her to do a tough abstract thing like giving a definition. She was worried. She looked around for Leo, and when she saw him she made that sign at him. He came bounding over and shoved Ramona away. Then he began to tell me how wise and good and gentle I am. He may be a genius, but even a genius chimp is still a chimp, and I told him I wasn't fooled by all his flattery. Then I asked *him* what the new sign meant.

"Jump high come again," Leo signed.

A simple chimp phrase referring to fun and frolic? So I thought at first, and so did many of my colleagues. But Dave Yost said, "Then why was Ramona so evasive about defining it?"

"Defining isn't easy for them," Beth Rankin said.

"Ramona's one of the five brightest. She's capable of it. Especially since the sign can be defined by use of four other established signs, as Leo proceeded to do."

"What are you getting at, Dave?" I asked.

Yost said, "*Jump high come again*' might be about a game they like to play, but it could also be an eschatological reference, sacred talk, a concise metaphorical way to speak of death and resurrection, no?"

Mick Falkenburg snorted. "Jesus, Dave, of all the nutty Jesuitical bullshit—"

"Is it?"

"It's possible sometimes to be too subtle in your analysis," Falkenburg said. "You're suggesting that these chimpanzees have a theology?"

"I'm suggesting that they may be in the process of evolving a religion," Yost replied.

Can it be?

Sometimes we lose our perspective with these animals, as Mick indicated, and we overestimate their intelligence; but just as often, I think, we underestimate them.

Jump high come again.

I wonder. Secret sacred talk? A chimpanzee theology? Belief in life after death? A religion?

* * *

They know that human beings have a body of ritual and belief that they call religion, though how much they really comprehend about it is hard to tell. Dave Yost, in his metaphysical discussions with Leo and some of the other alphas, introduced the concept long ago. He drew a hierarchy that began with God and ran downward through human beings and chimpanzees to dogs and cats and onward to insects and frogs, by way of giving the chimps some sense of the great chain of life. They had seen bugs and frogs and cats and dogs, but they wanted Dave to show them God, and he was forced to tell them that God is not actually tangible and accessible, but lives high overhead although His essence penetrates all things. I doubt that they grasped much of that. Leo, whose nimble and probing intelligence is a constant illumination to us, wanted Yost to explain how we talked to God and how God talked to us, if He wasn't around to make signs, and Yost said that we had a thing called religion, which was a system of communicating with God. And that was where he left it, a long while back.

Now we are on guard for any indications of a developing religious consciousness among our troop. Even the scoffers—Mick Falkenburg, Beth, to some degree, Charley Damiano—are paying close heed. After all, one of the underlying purposes of this project is to reach an understanding of how the first hominids managed to cross the intellectual boundary that we like to think separates the animals from humanity. We can't reconstruct a bunch of Australopithecines and study them; but we *can* watch chimpanzees who have been given the gift of language build a quasi-protohuman society, and it is the closest thing to traveling back in time that we are apt to achieve. Yost thinks, I think, Burt Christensen is beginning to think, that we have inadvertently kindled an awareness of the divine, of the numinous force that must be worshipped, by allowing them to see that their gods—us—can be struck down and slain by an even higher power.

The evidence so far is slim. The attention given Vendelmans and Judy; the solitary meditations of Leo and Grimsky; the large gatherings in the grove; the greatly accelerated use of modified sign language in chimp-to-chimp talk at those gatherings; the potentially eschatological reference we think we see in the sign that Leo translated as *jump high come again*. That's it. To those of us who want to interpret that as the foundations of religion, it seems indicative of

what we want to see; to the rest, it all looks like coincidence and fantasy. The problem is that we are dealing with non-human intelligence and we must take care not to impose our own thought-constructs. We can never be certain if we are operating from a value system anything like that of the chimps. The built-in ambiguities of the sign-language grammar we must use with them complicate the issue. Consider the phrase "banana fly" that Leo used in a speech—a sermon?—in the oak grove, and remember Ramona's reference to the sick Vendelmans as "rotten banana." If we take *fly* to be a verb, "banana fly" might be considered a metaphorical description of Vendelmans's ascent to heaven. If we take it to be a noun, Leo might have been talking about the *Drosophila* flies that feed on decaying fruit, a metaphor for the corruption of the flesh after death. On the other hand, he may simply have been making a comment about the current state of our garbage dump.

We have agreed for the moment not to engage the chimpanzees in any direct interrogation about any of this. The Heisenberg principle is eternally our rule here: the observer can too easily perturb the thing observed, so we must make only the most delicate of measurements. Even so, of course, our presence among the chimps is bound to have its impact, but we do what we can to minimize it by avoiding leading questions and watching in silence.

Two unusual things today. Taken each by each, they would be interesting without being significant; but if we use each to illuminate the other, we begin to see things in a strange new light, perhaps.

One thing is an increase in vocalizing, noticed by nearly everyone, among the chimps. We know that chimpanzees in the wild have a kind of rudimentary spoken language—a greeting-call, a defiance-call, the grunts that mean "I like the taste of this," the male chimp's territorial hoot, and such: nothing very complex, really not qualitatively much beyond the language of birds or dogs. They also have a fairly rich non-verbal language, a vocabulary of gestures and facial expressions; but it was not until the first experiments decades ago in teaching chimpanzees human sign language that any important linguistic capacity became apparent in them. Here at the research station the chimps communicate almost wholly in signs, as they have been trained to do for generations and as they have taught their young ones to do; they revert to hoots and grunts only in the most elemental situations. We ourselves communicate mainly in signs when we

are talking to each other while working with the chimps, and even in our humans-only conferences we use signs as much as speech, from long habit. But suddenly the chimps are making sounds at each other. Odd sounds, unfamiliar sounds, weird clumsy imitations, one might say, of human speech. Nothing that we can understand, naturally; the chimpanzee larynx is simply incapable of duplicating the phonemes humans use. But these new grunts, these tortured blurts of sound, seem intended to mimic our speech. It was Damiano who showed us, as we were watching a tape of a grove session, how Attila was twisting his lips with his hands in what appeared unmistakably to be an attempt to make human sounds come out.

Why?

The second thing is that Leo has started wearing a shirt and a hat. There is nothing remarkable about a chimp in clothing; although we have never encouraged such anthropomorphization here, various animals have taken a fancy from time to time to some item of clothing, have begged it from its owner, and have worn it for a few days or even weeks. The novelty here is that the shirt and the hat belonged to Hal Vendelmans, and that Leo wears them only when the chimps are gathered in the oak grove, which Dave Yost has lately begun calling the "holy grove." Leo found them in the toolshed beyond the vegetable garden. The shirt is ten sizes too big, Vendelmans having been so brawny, but Leo ties the sleeves across his chest and lets the rest dangle down over his back almost like a cloak.

What shall we make of this?

Jan is the specialist in chimp verbal processes. At the meeting to-night she said, "It sounds to me as if they're trying to duplicate the rhythms of human speech even though they can't reproduce the actual sounds. They're playing at being human."

"Talking the god-talk," said Dave Yost.

"What do you mean?" Jan asked.

"Chimps talk with their hands. Humans do too, when speaking with chimps, but when humans talk to humans they use their voices. Humans are gods to chimps, remember. Talking in the way the gods talk is one way of remaking yourself in the image of the gods, of putting on divine attributes."

"But that's nonsense," Jan said. "I can't possibly—"

"Wearing human clothing," I broke in excitedly, "would also be a kind of putting on divine attributes, in the most literal sense of the phrase. Especially if the clothes—"

"—had belonged to Hal Vendelmans," said Christensen.

"The dead god," Yost said.

We looked at each other in amazement.

Charley Damiano said, not in his usual skeptical way but in a kind of wonder, "Dave, are you hypothesizing that Leo functions as some sort of priest, that those are his sacred garments?"

"More than just a priest," Yost said. "A high priest, I think. A pope. The pope of the chimps."

Grimsky is suddenly looking very feeble. Yesterday we saw him moving slowly through the meadow by himself, making a long circuit of the grounds as far out as the pond and the little waterfall, then solemnly and ponderously staggering back to the meeting-place at the far side of the grove. Today he has been sitting quietly by the stream, occasionally rocking slowly back and forth, now and then dipping his feet in. I checked the records: he is 43 years old, well along for a chimp, although some have been known to live fifty years and more. Mick wanted to take him to the infirmary but we decided against it; if he is dying, and by all appearances he is, we ought to let him do it with dignity in his own way. Jan went down to the grove to visit him and reported that he shows no apparent signs of disease. His eyes are clear, his face feels cool. Age has withered him and his time is at hand. I feel an enormous sense of loss, for he has a keen intelligence, a long memory, a shrewd and thoughtful nature. He was the alpha male of the troop for many years, but a decade ago, when Leo came of age, Grimsky abdicated in his favor with no sign of a struggle. Behind Grimsky's grizzled forehead there must lie a wealth of subtle and mysterious perceptions, concepts, and insights about which we know practically nothing, and very soon all that will be lost. Let us hope he's managed to teach his wisdom to Leo and Attila and Alice and Ramona.

Today's oddity: a ritual distribution of meat.

Meat is not very important in the diet of chimps, but they do like to have some, and as far back as I can remember Wednesday has been meat day here, when we give them a side of beef or some slabs of mutton or something of that sort. The procedure for dividing up the meat betrays the chimps' wild heritage, for the alpha males eat their fill first, while the others watch, and then the weaker males beg for a share and are allowed to move in to grab, and finally the fe-

males and young ones get the scraps. Today was meat day. Leo, as usual, helped himself first, but what happened after that was astounding. He let Attila feed, and then told Attila to offer some meat to Grimsky, who is even weaker today and brushed it aside. *Then Leo put on Vendelmans's hat* and began to parcel out scraps of meat to the others. One by one they came up to him in the current order of ranking and went through the standard begging maneuver, hand beneath chin, palm upward, and Leo gave each one a strip of meat.

"Like taking communion," Charley Damiano muttered. "With Leo the celebrant at the Mass."

Unless our assumptions are totally off base, there is a real religion going on here, perhaps created by Grimsky and under Leo's governance. And Hal Vendelmans's faded old blue work-hat is the tiara of the pope.

Beth Rankin woke me at dawn and said, "Come fast. They're doing something strange with old Grimsky."

I was up and dressed and awake in a hurry. We have a closed-circuit system now that pipes the events in the grove back to us, and we paused at the screen so that I could see what was going on. Grimsky sat on his knees at the edge of the stream, eyes closed, barely moving. Leo, wearing the hat, was beside him, elaborately tying Vendelmans's shirt over Grimsky's shoulders. A dozen or more of the other adult chimps were squatting in a semicircle in front of them.

Burt Christensen said, "What's going on? Is Leo making Grimsky the assistant pope?"

"I think Leo is giving Grimsky the last rites," I said.

What else could it have been? Leo wore the sacred headdress. He spoke at length using the new signs—the ecclesiastical language, the chimpanzee equivalent of Latin or Hebrew or Sanskrit—and as his oration went on and on, the congregation replied periodically with outbursts of—I suppose—response and approval, some in signs, some with the grunting garbled pseudo-human sounds that Dave Yost thought was their version of god-talk. Throughout it all Grimsky was silent and remote, though occasionally he nodded or murmured or tapped both his shoulders in a gesture whose meaning was unknown to us. The ceremony went on for more than an hour. Then Grimsky leaned forward, and Kong and Chump took him by the arms and eased him down until he was lying with his cheek against the ground.

For two, three, five minutes all the chimpanzees were still. At last Leo came forward and removed his hat, setting it on the ground beside Grimsky, and with great delicacy he untied the shirt Grimsky wore. Grimsky did not move. Leo draped the shirt over his own shoulders and donned the hat again.

He turned to the watching chimps and signed, using the old signs that were completely intelligible to us, "Grimsky now be human being."

We stared at each other in awe and astonishment. A couple of us were sobbing. No one could speak.

The funeral ceremony seemed to be over. The chimps were dispersing. We saw Leo sauntering away, hat casually dangling from one hand, the shirt, in the other, trailing over the ground. Grimsky alone remained by the stream. We waited ten minutes and went down to the grove. Grimsky seemed to be sleeping very peacefully, but he was dead, and we gathered him up—Burt and I carried him; he seemed to weigh almost nothing—and took him back to the lab for the autopsy.

In mid-morning the sky darkened and lightning leaped across the hills to the north. There was a tremendous crack of thunder almost instantly and sudden tempestuous rain. Jan pointed to the meadow. The male chimps were doing a bizarre dance, roaring, swaying, slapping their feet against the ground, hammering their hands against the trunks of the trees, ripping off branches and flailing the earth with them. Grief? Terror? Joy at the translation of Grimsky to a divine state? Who could tell? I had never been frightened by our animals before—I knew them too well, I regarded them as little hairy cousins—but now they were terrifying creatures and this was a scene out of time's dawn, as Gonzo and Kong and Attila and Chump and Buster and Claudius and even Pope Leo himself went thrashing about in that horrendous rain, pounding out the steps of some unfathomable rite.

The lightning ceased and the rain moved southward as quickly as it had come, and the dancers went slinking away, each to his favorite tree. By noon the day was bright and warm and it was as though nothing out of the ordinary had happened.

Two days after Grimsky's death I was awakened again at dawn, this time by Mick Falkenburg. He shook my shoulder and yelled at me to wake up, and as I sat there blinking he said, "Chicory's dead!

I was out for an early walk and I found her near the place where Grimsky died."

"Chicory? But she's only—"

"Eleven, twelve, something like that. I know."

I put my clothes on while Mick woke the others, and we went down to the stream. Chicory was sprawled out, but not peacefully—there was a dribble of blood at the corner of her mouth, her eyes were wide and horrified, her hands were curled into frozen talons. All about her in the moist soil of the streambank were footprints. I searched my memory for an instance of murder in the chimp community and could find nothing remotely like it—quarrels, yes, and lengthy feuds, and some ugly ambushes and battles, fairly violent, serious injuries now and then. But this had no precedent.

"Ritual murder," Yost murmured.

"Or a sacrifice, perhaps?" suggested Beth Rankin.

"Whatever it is," I said, "they're learning too fast. Recapitulating the whole evolution of religion, including the worst parts of it. We'll have to talk to Leo."

"Is that wise?" Yost asked.

"Why not?"

"We've kept hands off so far. If we want to see how this thing unfolds—"

"During the night," I said, "the Pope and the College of Cardinals ganged up on a gentle young female chimp and killed her. Right now they may be off somewhere sending Alice or Ramona or Anna Livia's twins to chimp heaven. I think we have to weigh the value of observing the evolution of chimp religion against the cost of losing irreplaceable members of a unique community. I say we call in Leo and tell him that it's wrong to kill."

"He knows that," said Yost. "He must. Chimps aren't murderous animals."

"Chicory's dead."

"And if they see it as a holy deed?" Yost demanded.

"Then one by one we'll lose our animals, and at the end we'll just have a couple of very saintly survivors. Do you want that?"

We spoke with Leo. Chimps can be sly and they can be manipulative, but even the best of them, and Leo is the Einstein of chimpanzees, does not seem to know how to lie. We asked him where Chicory was and Leo told us that Chicory was now a human being. I

felt a chill at that. Grimsky was also a human being, said Leo. We asked him how he knew that they had become human and he said, "They go where Vendelmans go. When human go away, he become god. When chimpanzee go away, he become human. Right?"

"No," we said.

The logic of the ape is not easy to refute. We told him that death comes to all living creatures, that it is natural and holy, but that only God could decide when it was going to happen. God, we said, calls His creatures to Himself one at a time. God had called Hal Vendelmans, God had called Grimsky, God would someday call Leo and all the rest here. But God had not yet called Chicory. Leo wanted to know what was wrong with sending Chicory to Him ahead of time. Did that not improve Chicory's condition? No, we replied. No, it only did harm to Chicory. Chicory would have been much happier living here with us than going to God so soon. Leo did not seem convinced. Chicory, he said, now could talk words with her mouth and wore shoes on her feet. He envied Chicory very much.

We told him that God would be angry if any more chimpanzees died. We told him that we would be angry. Killing chimpanzees was wrong, we said. It was not what God wanted Leo to be doing.

"Me talk to God, find out what God wants," Leo said.

We found Buster dead by the edge of the pond this morning, with indications of another ritual murder. Leo coolly stared us down and explained that God had given orders that all chimpanzees were to become human beings as quickly as possible, and this could only be achieved by the means employed on Chicory and Buster.

Leo is confined now in the punishment tank and we have suspended this week's meat distribution. Yost voted against both of those decisions, saying we ran the risk of giving Leo the aura of a religious martyr, which would enhance his already considerable power. But these killings have to stop. Leo knows, of course, that we are upset about them. But if he believes his path is the path of righteousness, nothing we say or do is going to change his mind.

Judy Vendelmans called today. She has put Hal's death fairly well behind her, misses the project, misses the chimps. As gently as I could, I told her what has been going on here. She was silent a very long time—Chicory was one of her favorites, and Judy has had enough grief already to handle for one summer—but finally she said,

"I think I know what can be done. I'll be on the noon flight tomorrow."

We found Mimsy dead in the usual way late this afternoon. Leo is still in the punishment tank—the third day. The congregation has found a way to carry out its rites without its leader. Mimsy's death has left me stunned, but we are all deeply affected, virtually unable to proceed with our work. It may be necessary to break up the community entirely to save the animals. Perhaps we can send them to other research centers for a few months, three of them here, five there, until this thing subsides. But what if it doesn't subside? What if the dispersed animals convert others elsewhere to the creed of Leo?

The first thing Judy said when she arrived was, "Let Leo out. I want to talk with him."

We opened the tank. Leo stepped forth, uneasy, abashed, shading his eyes against the strong light. He glanced at me, at Yost, at Jan, as if wondering which one of us was going to scold him; and then he saw Judy and it was as though he had seen a ghost. He made a hollow rasping sound deep in his throat and backed away. Judy signed hello and stretched out her arms to him. Leo trembled. He was terrified. There was nothing unusual about one of us going on leave and returning after a month or two, but Leo must not have expected Judy ever to return, must in fact have imagined her gone to the same place her husband had gone, and the sight of her shook him. Judy understood all that, obviously, for she quickly made powerful use of it, signing to Leo, "I bring you message from Vendelmans."

"Tell tell tell!"

"Come walk with me," said Judy.

She took him by the hand and led him gently out of the punishment area and into the compound, and down the hill toward the meadow. I watched from the top of the hill, the tall slender woman and the compact, muscular chimpanzee close together, side by side, hand in hand, pausing now to talk, Judy signing and Leo replying in a flurry of gestures, then Judy again for a long time, a brief response from Leo, another cascade of signs from Judy, then Leo squatting, tugging at blades of grass, shaking his head, clapping hand to elbow in his expression of confusion, then to his chin, then taking Judy's hand. They were gone for nearly an hour. The other chimps did not dare approach them. Finally Judy and Leo, hand in hand, came

quietly up the hill to headquarters again. Leo's eyes were shining and so were Judy's.

She said, "Everything will be all right now. That's so, isn't it, Leo?"

Leo said, "God is always right."

She made a dismissal sign and Leo went slowly down the hill. The moment he was out of sight, Judy turned away from us and cried a little, just a little; then she asked for a drink; and then she said, "It isn't easy, being God's messenger."

"What did you tell him?" I asked.

"That I had been in heaven visiting Hal. That Hal was looking down all the time and he was very proud of Leo, except for one thing: that Leo was sending too many chimpanzees to God too soon. I told him that God was not yet ready to receive Chicory and Buster and Mimsy, that they would have to be kept in storage cells for a long time, until their true time came, and that that was not good for them. I told him that Hal wanted Leo to know that God hoped he would stop sending him chimpanzees. Then I gave Leo Hal's old wristwatch to wear when he conducts services, and Leo promised he would obey Hal's wishes. That was all. I suspect I've added a whole new layer of mythology to what's developing here, and I trust you won't be angry with me for doing it. I don't believe any more chimps will be killed. And I think I'd like another drink."

Later in the day we saw the chimps assembled by the stream. Leo held his arm aloft and sunlight blazed from the band of gold on his slim hairy wrist, and a great outcry of grunts in god-talk went up from the congregation and they danced before him, and then he donned the sacred hat and the sacred shirt and moved his arms eloquently in the secret sacred gestures of the holy sign language.

There have been no more killings. I think no more will occur. Perhaps after a time our chimps will lose interest in being religious, and go on to other pastimes. But not yet, not yet. The ceremonies continue, and grow ever more elaborate, and we are compiling volumes of extraordinary observations, and God looks down and is pleased. And Leo proudly wears the emblems of his papacy as he bestows his blessing on the worshippers in the holy grove.

THE VITANULS

John Brunner

Medical advances, especially in the field of infant mortality, have already given us one serious ethical problem: over-population. But imagine how that problem could be exacerbated should medical research discover a way substantially to extend our lives—a cure for senility, perhaps. Suppose, for instance, that the earth's population grew to number as many people alive at one time as had ever lived before in man's history. British science-fiction author John Brunner examines that proposition in an unusual but appropriate setting—a maternity hospital in India. The setting and a remarkably spiritual central character provide the story with some considerations of life and death not frequently found in science fiction, nor expressed, as they are here, in Hindu values. Yet Brunner weaves his tale so sympathetically that even to the Western mind, the reaction of a great Indian physician to the crisis seems altogether reasonable and fitting.

Before the soundproof, germproof double glass window of the delivery room the matron of the maternity hospital came to a halt. "And there," she told the tall young American from the World Health Organisation, "is our patron saint."

Barry Chance blinked at her. She was a brisk fortyish Kashmiri woman with an aura of efficiency, not at all the kind of person one would expect to make jokes about her life's work. And indeed there had been no trace of jocularly in her tone. But in this teeming subcontinent of India a stranger could never be sure.

He compromised. "I'm sorry," he lied. "I didn't catch that . . . ?"

Out of the corner of his eye he studied the man the matron had indicated. He was elderly and balding; what little hair remained to him had whitened into a sort of halo around his heavily lined face. Most Indians, the American had noticed, tended to grow fat with age, but this man had gone scrawny, like Gandhi. Surely, though, an ascetic appearance and a halo of hair weren't enough to establish a claim to sainthood?

"Our patron saint," the matron repeated, sublimely unaware of her visitor's bewilderment. "Dr. Ananda Kotiwala. You're very fortunate to see him at work. It's his last day here before he retires."

Struggling to make sense of her remarks, Chance stared unashamedly at the old man. He felt his rudeness was excused by the fact that this corridor adjoining the delivery room was a kind of public gallery. On every side there were relatives and friends of the expectant mothers, down to and including very small children who had to stand on tiptoe to peer in through the double glass window. There was no such thing as privacy in India unless you were rich, and in any overcrowded, underdeveloped country a minimal fraction of the people enjoyed that luxury he'd taken for granted since childhood.

The fact that toddlers could watch, fascinated, the arrival of their new brothers and sisters was accepted here as a part of growing up. Chance reminded himself sternly that he was a foreigner, and—what was more—a doctor himself, trained at one of the few colleges that still administered the Hippocratic Oath in full form to its graduates. He forced his personal preconceptions to the back of his mind and concentrated on unravelling the curious comment the matron had made.

The scene before him offered no hints. All he could see was a typical Indian hospital delivery room, containing thirty-six mothers in labour, of whom two were suffering agonies and screaming—at least, to judge by their open mouths; the soundproofing was extremely good.

He wondered briefly how Indians felt about their children entering the world under such circumstances. What it suggested to him was an assembly-line, the mothers reduced to machines producing their quota of infants ac-

according to a preplanned schedule. And all of it so dreadfully public!

Again, though, he was falling into the trap of thinking like a modern American, parochially. For untold generations most of mankind had been born publicly. Although it had been estimated that the world's present living population was just about equal in number to all the human beings there had ever been before the twenty-first century, the majority of Earth's people continued that ancient tradition, and made a birth a social event: in villages, an excuse for a celebration, or as here a sort of family outing to the hospital.

The modern aspects of the event were easily listed. The attitudes of the mothers, for instance: one could tell at a glance which of them had enjoyed up-to-date parental instruction, for their eyes were closed and their faces bore expressions of determination. They knew what miracle was going on in their bodies, and they intended to help it, not resist it. Good. Chance nodded approval. But there remained the women who were screaming, as much from terror as pain, probably . . .

He shifted his attention with an effort. After all, he was supposed to be conducting a survey of the methods in use here.

The latest recommendations of the experts seemed to be being properly applied—you'd expect as much in a large city where most of the medical staff had had the benefit of training abroad. Some time soon he was committed to going out into the villages, and things would be different there, but he'd think about that when he had to.

The elderly doctor who had been called "our patron saint" was just completing the delivery of a boy. Gloved hand held up the latest recruit to the army of humanity, glistening. A slap—correction: a beautifully restrained tap with the open palm, enough to provoke a squall and the first deep breath, but not enough to aggravate the birth-trauma. And handed over to the waiting nurse to be laid on the little bench beside the bed, lower than the mother so that the last few precious ounces of maternal blood could seep down from the placenta before the cord was severed.

Excellent. All in accordance with the best modern prac-

tice. Except—why was the doctor having to explain so much so patiently to the awkward girl holding the child?

Chance's puzzlement was brief; then he remembered. Of course. There weren't enough trained nurses in this country to allot one to every new mother. So those girls standing neat and scared in their disposable plastic overalls, their lank black hair bound in sterile plastic snoods, would be younger sisters or eldest daughters doing their best to help out.

Then the old man, with a final smile of reassurance, was leaving the worried girl and going to hold the hand of one of the women screaming.

Chance watched with satisfaction as he soothed her, bringing about a complete relaxation within moments, and—as far as could be guessed through the double barrier of soundproof glass and incomprehensible language—instructing her how best to hasten the delivery. Yet there was nothing more here than he'd seen in a hundred hospitals.

He turned to the matron and asked directly, "Why do you call him 'patron saint'?"

"Dr. Kotiwala," the matron said, "is the most—now what would it be in English? Is there the word 'empathetic'?"

"From 'empathy'?" Chance frowned. "I don't think so. But I get what you mean, anyway."

"Yes," the matron said. "Did you not see how he quieted the one who was screaming?"

Chance gave a slow nod. Yes, come to think of it, it must be regarded as a special gift in a country like this—to be able to break through the superstitious fright of a woman barely above peasant level and make her see what it had taken other women around her the full nine months of pregnancy and much skilled instruction to understand. Now there was only one woman screaming with her mouth wide, and the doctor was soothing her in her turn. The other he'd just spoken to was struggling to encourage her contractions.

"Dr. Kotiwala is wonderful," the matron went on. "Everybody loves him. I have known parents to consult astrologers not to determine the most fortunate birthday for their child but only how to make sure it is born during Dr. Kotiwala's shift in the delivery room."

Shift? Oh yes: they operated a three-shift day. Once

more the image of the assembly-line came to him. But it was far too advanced a concept to reconcile with the idea of applying to astrologers. What a crazy country! Chance repressed a shiver and admitted to himself that he'd be glad when he was allowed to return home.

For long moments after that he was silent, noticing something he hadn't previously spotted: how, when the labour pains permitted, the mothers opened their eyes and hopefully followed Dr. Kotiwala's progress around the room as if wanting to invite him to spend a minute or two at their bedside.

This time their hopes weren't fulfilled. There was a breech presentation at the far end, and it would take careful manipulation to reverse the baby. Plastic-clad, a beautiful dark girl of about fifteen bent to watch what the doctor was doing, putting out her right hand so that the tense anxious mother could clasp it for comfort.

By his own standards, Chance thought, there was nothing extraordinary about Kotiwala. He was obviously competent, and his patients appeared to like him, but he was old and rather slow, and one could see how cautiously he moved now the end of his shift was near and he was tiring.

On the other hand, it was admirable to find the human touch in a birth-factory like this. He'd asked the matron, within minutes of his arrival, how long the average stay of a patient might be, and she'd said with a wry smile, "Oh—twenty-four hours for the easy ones, and perhaps thirty-six for cases with complications."

Looking at Dr. Kotiwala, one might have assumed there was all the time in the world.

From an American standpoint even that didn't constitute a claim to sanctity, but through Indian eyes doubtless things looked different. The matron had warned him that he'd come at a busy time, nine months after a big spring religious festival which people regarded as auspicious for increasing their families. The warning hadn't prepared him for the reality; the hospital was packed.

Yet it could have been worse. He shuddered a little. The back of the problem was broken, but there were still something like 180,000 new mouths to feed every day. At the peak of the population explosion there had been just under a quarter-million per day; then the impact of

modern medicine was felt, even people in India, China and Africa began to recognise the need to plan for the children they could afford to feed, clothe and educate, and the crisis diminished.

Nonetheless it would be years before the children of that peak period could become teachers, workers, doctors to cope with the tremendous pressure. Thinking along these lines brought him to a subject which had been engaging a lot of his attention recently, and he spoke aloud without intending to.

"People like him, in this of all jobs—that's who they ought to choose!"

"I beg your pardon?" the matron said with positively British formality. The British had left ineradicable traces on the intellectuals of this country.

"Nothing," Chance muttered.

"But did you not say someone ought to choose Dr. Kotiwala for something?"

Annoyed with himself, yet—once reminded of the dilemma shortly to be sprung on the world—unable to hold his tongue, Chance gave ground.

"You said it was Dr. Kotiwala's last day here, didn't you?"

"Why, yes. He retires tomorrow."

"You have someone in mind to replace him?"

The matron shook her head vigorously. "Oh no! In the physical sense, yes, in that another doctor will take his shifts, but men like Dr. Kotiwala are rare in any generation and in modern times most of all. We're dreadfully sorry to lose him."

"Has he—uh—passed some arbitrary retirement age?"

The matron smiled thinly. "Hardly, in India! We can't afford the luxuries you Americans go in for, and that includes putting usable material—human or otherwise—on a scrapheap before it's worn out."

With his eyes on the elderly doctor, who had successfully reversed the breech presentation and moved on to attend to the woman in the next bed, Chance said, "He's retiring voluntarily, then."

"Yes."

"Why? Has he lost interest in his work?"

The matron was shocked. "Of course not! I'm not sure I can make his reasons clear, though." She bit her lip.

"Well, he is very old now, and he fears that one day soon a child may die because he has let his attention wander. It would set him back many steps on the road to enlightenment if that happened."

Chance felt a surge of enlightenment himself. Believing he completely understood what the matron meant, he said, "In that case he damned well deserves—"

And broke off, because strictly he ought not to be thinking about this subject, or talking about it.

"What?" the matron said, and when Chance shook his head, went on: "You see, when he was young Dr. Kotiwala was much influenced by the teachings of the Jains, to whom the taking of any life at all is repugnant. When his desire to cherish life led him to study as a doctor, he had to accept that some killing—of bacteria, for example—is inevitable to ensure human survival. His kindness is based on religious principles. And it would be more than he could bear to think that his own arrogance in continuing to work when it was no longer safe had cost an innocent baby its life."

"He can hardly be a Jain now," Chance said, lacking anything else by way of comment. Privately he was thinking that if what the matron said was correct, there were some old fossils back home who could do with a dose of Kotiwala's humility instead of hanging on until they were practically senile.

"He's a Hindu, as are most of our people," the matron explained. "Though he tells me his thinking has been much influenced by the teachings of Buddhism—which began, after all, as a Hindu heresy." She didn't sound greatly interested in what she was saying. "But I'm afraid I still don't understand to what you were referring a moment ago," she added.

Chance thought of gigantic factories owned by Du Pont, Bayer, Glaxo and heaven knows who else, labouring night and day with more expenditure of energy than a million mothers bringing forth commonplace human beings, and decided that the facts were going to be public knowledge soon enough for him to risk lifting a corner of the curtain of secrecy. It was getting him down to keep his mouth shut all the time.

He said, "Well, what I mean was that if I have any say in the matter people like him would get priority when

it comes to—uh—the most advanced kinds of medical treatment. To preserve somebody who's liked and admired seems better than saving someone who is mainly feared."

There was a pause.

"I think I follow you," the matron said wisely. "I take it that the anti-death pill is a success?"

Chance started. She gave him another of her wry smiles.

"Oh, it's difficult for us to keep up with the literature when we work under such pressure, but there have been hints, haven't there? You in the rich countries like America and Russia have been trying for years to find a broad-spectrum specific against aging, and I think—knowing your countries by hearsay—there must have been a long angry argument over who should benefit first."

Chance surrendered completely and gave a miserable nod. "Yes, there's a specific against senility. It isn't perfect, but pressure on the drug companies to put it into commercial production had grown so great that just before I left WHO Headquarters to come here I heard the contracts were being placed. A course of treatment will cost five or six hundred dollars and last for eight to ten years. I don't have to tell you what it'll mean. But if I had my way, I'd pick someone like your Dr. Kotiwala to enjoy the results before all the stupid old men with power and wealth who are going to have their obsolete ideas carried into the future by this break-through!"

He stopped short, alarmed at his own vehemence and hoping that none of the curious spectators surrounding them could speak English.

"Your attitude does you credit," the matron said. "But in one sense it's inexact to say Dr. Kotiwala is retiring. He might prefer to say he's changing his career. And if you offered to give him anti-senility treatment I expect he would smile and refuse."

"Why in the world—?"

"It is hard to make clear in English." The matron frowned. "You know what is a sunnyasi, perhaps?"

Bewildered, Chance said, "One of those holy men I've seen around the place, wearing nothing but a loincloth and carrying a begging-bowl."

"And a staff, usually."

"A sort of fakir?"

"Not in the least. A sunnyasi is a man in the final

stage of his life's work. He can have been anything previously—a businessman, commonly, or a civil servant, or a lawyer, or even a doctor."

"You mean your Dr. Kotiwala is going to throw away all his medical skills, all the service he could still perform in this overcrowded country even if he did risk the life of a baby one of these days, and go out begging in a loincloth for the sake of his own salvation?"

"This is why we call him our patron saint," the matron said with an affectionate smile in Dr. Kotiwala's direction. "When he is gone from here, and has collected such virtue, he will be a friend for us who remain behind."

Chance was appalled. A moment ago the matron had been saying India could not afford to throw aside people with good work still before them; now here she was seeming to approve a plan that struck him as compounded of equal parts selfishness and superstition.

"Are you telling me he believes this nonsense about stacking up virtue for a future existence?"

The matron gave him a cold stare. "I think that is uncivil of you. The teaching of Hinduism is that the soul is born again, throughout an eternal cycle, until it achieves one-ness with the All. Can you not appreciate how a lifetime of work among the newly-born makes all this real to us?"

"You believe it too?"

"That's irrelevant. But . . . I do witness miracles every time I admit a mother to this hospital. I witness how an animal act, a process with slimy, messy, *bloody* associations, brings about the growth of a reasoning being. I was born, and you, a squally helpless infant, and here we stand talking in abstract concepts. Maybe it's a mere function of chemical complexity; I don't know. I told you, I find it hard to keep up with the literature."

Chance stared through the window of the delivery room with a puzzled frown. He felt somehow disappointed—even cheated—after his near-acceptance of Dr. Kotiwala in the matron's admiring terms. At length he muttered, "I guess maybe we'd better move on."

The sensation of which Dr. Kotiwala was chiefly aware was weariness. It went all through his body, to the marrow of his bones.

There was no hint of it in his outward behaviour—no suggestion that he was mechanically going through the motions. The mothers who entrusted themselves and their offspring to his care would have detected any such failing, with perceptions deeper than ordinary, and he would have known the truth himself and felt he was betraying his trust.

But he was unspeakably, incredibly tired.

More than sixty years had passed since he graduated from medical school. There had been no change in the way human beings were created. Oh, the trappings had changed as medicine made its successive impacts; he remembered the inarguable disasters caused by drugs like thalidomide, and the upside-down blessings of antibiotics that swamped countries like his own with more mouths than it could possibly feed, and now he was working with techniques which meant that nine out of ten of the children born under his supervision were wished-for, loved by their parents instead of being a burden or condemned to the half-life of illegitimacy.

Sometimes things turned out well, and sometimes badly. In the course of his long and valuable career Dr. Kotiwala had come to place reliance on no other principle.

Tomorrow . . .

His mind threatened to wander away from what he was doing: bringing to independent life the latest of all those he had delivered. How many thousands of mothers had moaned on the bed before him? He dared not count. And how many more thousands of new lives had he helped to launch? Those he less-than-dared to count. Perhaps he'd introduced to life a thief, an ingrate, a murderer, a fratricide . . .

No matter. Tomorrow—indeed, today, for his shift was over and this baby he was now raising by the feet was the last he would ever deliver in a hospital, though if he were appealed to in some miserable village he would doubtless help . . . Tomorrow there would be an end to worldly attachments. He would commit himself to the life of the spirit, and—

He checked. The woman alongside the mother, her sister-in-law, very much disturbed by the things she had had to do, like sterilising her hands in disinfectant and stripping

off her best sari in favour of a clammy plastic overall, spoke a fearful question.

He hesitated over his answer. To the superficial glance nothing about the baby seemed amiss. It was a boy, physically whole, the usual flushed postnatal colour, letting out an acceptable scream to greet the world. All was as it should be. And yet . . .

He cradled the baby on his left arm while deftly raising first one, then the other eyelid. Sixty years of practice had made him gentle. He stared deep into the vacant light eyes, contrasting almost frighteningly with the skin around them.

Beyond them was—was . . .

But what could one say to a child as new as this? He sighed and gave it into the care of the sister-in-law, and the clock on the wall ticked away the last few seconds of his shift.

Nonetheless his mind remained on the indefinable impulse which had compelled him to take a second look at the boy. When the doctor taking over from him arrived, Dr. Kotiwala finished his summary briefing by saying, "And there's something wrong with the boy just born in Bed 32. I can't put my finger on it. But if you get the chance, check him over, would you?"

"Will do," said the relief doctor, a fat young man from Benares with a shiny brown face and shiny soft hands.

The matter continued to irk Dr. Kotiwala even though he'd spoken about it. Changed, showered, ready to leave, he still lingered in the corridor to watch his colleague checking the baby as requested, making a thorough inspection from head to sole. He found nothing wrong, and catching sight of Dr. Kotiwala as he turned away, spread his hands and shrugged, his attitude implying, "Fuss about nothing if you ask me!"

Yet when I looked into those eyes, something behind them suggested . . .

No, it was absurd. What could any adult hope to read in the eyes of a brand-new baby? Wasn't it a kind of arrogance that made him think his colleague was missing something of vital importance? In a dilemma, he considered the idea of going back in the delivery room and taking another look.

"Isn't that your patron saint standing there?" Chance muttered to the matron in a cynical tone.

"Why, so it is. How fortunate! Now you can make his acquaintance yourself—if you wish to."

"You've painted him in such glowing colours," Chance said dryly, "I feel I'd be wasting a chance if I didn't meet him before he takes off his clothes and goes native."

The irony was almost completely lost on the matron. She bustled ahead with exclamations, but interrupted herself the moment she noticed Kotiwala's glum expression.

"Doctor! Is something wrong?"

"I don't know," Kotiwala sighed. His English was good, but heavily accented in the singsong rhythm which the departed British rulers of India had nicknamed "Bombay Welsh." It is the child just born in Bed 32, a boy. I am sure something is wrong, but as for what it is, I'm lost."

"In that case we must see to him," the matron said briskly. Clearly she had implicit faith in Kotiwala's opinions.

"Dr. Banerji has examined him and doesn't agree with me," Kotiwala countered.

In the matron's view Kotiwala was Kotiwala and Banerji was nobody; her expression said so, louder than words. It struck Chance that here was his opportunity to find out whether the matron's trust had any real basis.

"Look, rather than taking up Dr. Banerji's time—he has a lot to cope with in there—why don't you bring the child out and we'll take a look at him?"

"Dr. Chance, from WHO," the matron explained. Absently Dr. Kotiwala shook hands.

"Yes, that is a good idea. A second opinion, wouldn't you say?"

It had been in the back of Chance's mind that his comparatively fresh training would enable him to apply some tests Kotiwala wasn't accustomed to using. In fact it was the other way around; the slow, thorough palpating of the child's body and limbs with which the old Indian began wasn't in Chance's line at all. Of course, it had its advantages, provided one knew the normal location of every bone and major muscle in the infantile frame. Anyway, it revealed nothing.

Heart normal, blood pressure average, all external ap-

pearances healthy, reflexes present and vigorous, fontanelle a trifle large but within normal range of variation . . .

After nearly three-quarters of an hour, Chance was convinced the old man was doing it to make an impression, and consequently was losing his temper by degrees. He noticed that Kotiwala again and again rolled up the boy's eyelids and stared into the eyes, as if he could read the brain behind. On the latest repetition of the act he snapped, "Tell me, doctor! What do you see in his eyes, hm?"

"What do you see?" Kotiwala countered, and motioned for Chance to look also.

"Nothing," Chance grunted a moment later. They'd checked the eyes, hadn't they, along with everything else?

"That's what I see, also," Kotiwala said. "Nothing."

Oh, for goodness' sake! Chance spun on his heel and went to the side of the room, peeling off his gloves preparatory to dumping them in the disposer. Over his shoulder he said, "Frankly, I don't see anything wrong with the kid at all. What do you think it is? The soul of an earthworm turned up in his body by mistake, or something?"

Kotiwala could hardly have missed the scorn with which the words were uttered, but his reply was perfectly calm and civil.

"No, Dr. Chance, I think that hardly likely. After a great deal of contemplation I've come to the conclusion that the traditional ideas are inaccurate. The human condition is a human thing. It embraces the imbecile and the genius, but it doesn't overlap with any other species. Who could claim that the soul of a monkey, or a dog, is inferior to that which looks out of the dirty windows of a moron's eyes?"

"I certainly wouldn't," Chance said with sarcasm, and began to strip off his gown. Kotiwala sighed, and shrugged, and was silent.

Later . . .

The sunnyasi Ananda Bhagat wore nothing more than a loincloth, owned nothing more in all the world than the begging-bowl and staff he carried. Around him—for it was cold in the hill-country this bleak December—the people of the village shivered in their cheap coarse clothes,

spending as much time as they could spare huddled over their tiny fires. They burned woodchips, rarely charcoal, and even now a great deal of cowdung. The foreign experts had told them to use cowdung for fertiliser, but the warmth of a fire was closer to the present than the mystery of fixed nitrogen and next year's crops.

Ignoring the chill, ignoring the strong smoke of the fire as it wandered upward and filled the gloomy hut, Ananda Bhagat spoke soothingly to a fearful girl of about seventeen at whose breast a baby clung. He had looked into its eyes, and there he had seen—nothing.

It was not the first such in this village; it was not the first village where he had seen the same. He accepted that as a fact of existence. With the abandonment of the name Kotiwala had gone the preconceptions of the Doctor of Medicine, Trinity College, Dublin, who had obeyed the behests of intellectualism in the sterile wards of a big city hospital. Throughout his eighty-five years he had sensed a greater reality looming over him, and his final decision had been to commit himself to it.

Now, as he gazed wonderingly into the empty face of the child, he heard a noise. The young mother heard it also, and cowered because it was loud and growing louder. So far had Ananda Bhagat come from his former world that he had to make a conscious effort before he identified it. A drone in the sky. A helicopter, a rarity here; why should a helicopter come to any particular one of India's seventy thousand villages?

The young mother whimpered. "Be still, my daughter," the sunnyasi said. "I will go and see what it is."

He let her hand fall with a final comforting pat and went out of the misshapen doorway to stand on the cold windy street. The village had only one street. Shading his eyes with his thin hand, he peered upward into the sky.

Yes: a helicopter, circling and glinting in the weak winter sunlight. It was descending, but that was nothing to do with his emergence into plain view. Before he recognised the sound of it, it must already have been coming down.

He waited.

In a little while, the people came chattering out of their homes, wondering why the attention of the outer world in the form of this curious noisy machine should be turned

on them. Seeing that their marvellous visitor, the holy man, the sunnyasi—such as he were few these days, and to be cherished—was standing firm, they drew courage from his example and likewise stood up boldly.

The helicopter settled in a blast of whirling dust, a little away from the beaten track called a street, and a man jumped down from it: a tall fair-skinned foreigner. He looked the scene over slowly, spotted the sunnyasi and let out an exclamation. Calling something to his companions, he began to stride up the street. Two others came down to stand beside the machine and talk in low tones: a girl of about twenty, in a sari of blue and green, and a young man in flying overalls, the pilot.

Clutching her baby to her, the young mother also came out to see what was going on, her first child—a toddler—pursuing her on unsteady feet with a hand outstretched to catch at her sari if his balance failed him.

"Dr. Kotiwala!" the young man from the helicopter said.

"I was," the sunnyasi agreed in a rusty voice. The whole vocabulary of English had sloughed off his mind like a snake's overtight skin.

"For God's sake!" The young man's voice was harsh. "We've had enough trouble finding you without your playing word-games with us now we're here! Thirteen villages we've had to stop at on the way, picking up clues and being told you were here yesterday and moved on . . ."

He wiped his face with the back of his hand.

"My name's Barry Chance, in case you've forgotten me. We met at the hospital in—"

The sunnyasi interrupted. "I remember very well, thank you. But who am I, that you spend so much time and energy trying to find me?"

"As far as we can tell, you're the first person ever to have recognised a Vitnul."

There was a silence. During it, Chance could almost see the sunnyasi's persona fading, that of Dr. Kotiwala replacing it. The change was reflected in the voice, which resumed its "Bombay Welsh" rhythm on the next words.

"My Latin is negligible, for I only learned what was necessary in medicine, but I take it that would be from *vita*, a life, and *nullus* . . . You mean: like this one here?" He gestured for the young mother beside him to advance

a pace, and rested his hand lightly on her baby's back.

Chance looked the infant over and at length shrugged. "If you say so," he muttered. "She's only about two months old, isn't she? So without tests . . ."

His voice trailed briefly away.

"Yes, without tests!" he burst out abruptly. "That's the point! Do you know what became of the boy you said had something wrong with him, the very last one you delivered before you—ah—*retired*?" There was great fierceness in his voice, but it was not directed at the old man he was talking to. It was simply the outward sign that he had been driven to the end of his resources.

"I have seen many others since," Kotiwala said. It was definitely not the sunnyasi speaking now, but the trained doctor with a lifetime's experience behind him. "I can imagine, but tell me anyhow."

Chance gave him a look that reflected something close to awe. The inquisitive villagers gathered nearby recognised the expression, and deduced—though not even the best-educated among them could follow the rapid English words—that the stranger from the sky was affected by the aura of their holy man. They relaxed noticeably.

"Well . . . your friend the matron kept insisting that if you'd said something was wrong with him there *must* be something wrong, although I'd said he was okay, Dr. Banerji had said he was okay. She went on and on about it until it was interfering with my work and delaying my departure. So I said to hell with it and had him taken out to WHO in Delhi for the most complete battery of tests they could lay on. Can you guess what they found?"

Kotiwala rubbed his forehead wearily. "Total suppression of the alpha and theta rhythms?" he suggested.

"You did know!" The accusation in Chance's voice was enough to shatter the barrier of language and communicate to the listening villagers, some of whom stepped menacingly closer to the sunnyasi as though to defend him if they had to.

Kotiwala waved them back with a reassuring gesture. He said, "No, I didn't know. It just now came to me what you would find."

"Then how in heaven's name did you—?"

"How did I guess the boy wasn't normal? I can't explain that to you, Dr. Chance. It would take sixty years

of work in a maternity hospital, seeing scores of babies born every day, to make you see what I saw!"

Chance bit back a fierce retort and let his shoulders droop. "I'll have to swallow that. But the fact remains: you did realise, within minutes of the kid being born, even though he looked healthy and none of our tests has ever revealed any organic deformity, that his brain was—was empty and there was no mind in it! Christ, the job I had convincing them at WHO that you'd really done it, and the weeks of argument before they'd let me come back to India and try to track you down!"

"Your tests," Kotiwala said, as though the last sentence hadn't been spoken. "Many of them?"

Chance threw his hands in the air. "Doctor, where the hell have you been these past two years?"

"Walking barefoot from small village to small village," Kotiwala said, deliberately taking the question literally. "I haven't followed news from the world outside. This is a world for these people." He indicated the rough street, the mean shacks, the ploughed and planted fields, and the blue mountains closing all of it in.

Chance took a deep breath. "So you don't know and don't really care. Let me fill you in. Only a matter of weeks after I saw you, the news broke which led to my recall from India: reports of a sudden appalling rise in congenital imbecility. Normally a child begins to react in at least a sketch for a human pattern while it's still very young. Precocious kids smile quickly, and any kid is likely to distinguish movement and bright colours, and reach out to grasp things . . ."

"Except these you have named Vitanuls?"

"Exactly." Chance clenched his fists as though trying to seize something out of the air. "No life! None of the normal reactions! Absence of normal cerebral waves when you test them on the EEG, as though everything that makes a person human had been—had been left out!"

He levelled a challenging arm at Kotiwala's chest. "And you recognised the very first one of all! Tell me how!"

"Patience." Bowed by the weight of his years, Kotiwala still held himself with immense dignity. "This rise in imbecility—it struck you directly after I retired from the hospital?"

"No, of course not."

"Why 'of course'?"

"We were too tied up with—Oh, you've been out of touch, haven't you?" Chance spoke with bitter sarcasm.

"A minor triumph of medicine was making all the headlines, and giving WHO enough headaches to be going on with. The anti-senility treatment had been made public a few days after I saw you, and everybody and his uncle was standing on line yelling for it."

"I see," Kotiwala said, and his aged shoulders finally hunched into an attitude of despair.

"You see? What's that supposed to mean?"

"Forgive my interrupting. Continue, please."

Chance shivered, apparently as much at what he was remembering as at the chill December air. "We'd done our best, and postponed announcement of the treatment until there were stocks enough to treat several million applicants, but of course that was as bad as announcing it in the lab stage, because everybody's grandmother seemed to have died last Friday and here were people screaming at us for killing them by neglect, and—hell, you get the picture. Whichever way we handled it we came out wrong.

"And then shoveled in on top of that mess came the new one. Congenital imbecility hits ten per cent of birth, twenty, thirty! What's going on? Everybody whizzes around in little circles because just as we were congratulating ourselves on sorting out the row about the anti-senility treatment, here comes the most fantastic crisis in history and it's not going to break, it's going to get worse, and worse, and worse . . . Over the past two weeks the rate has topped eighty per cent. Do you understand that, or are you so sunk in your superstitions it doesn't bother you any longer? Out of every ten children born last week, no matter in what country or continent, eight are *mindless animals!*"

"And you think the one we examined together was the very first?" Kotiwala disregarded the harshness of the younger man's words; his eyes were staring, unfocused, into the blue distance over the mountains.

"As far as we can work it out." Chance spread his hands. "At any rate, when we checked back we found the first kids of which this had been reported had been born on that particular day, and I happened to remember

that the time of birth of the earliest we'd heard about was an hour or so after I met you."

"What happened on that day?"

"Nothing that could account for it. Every resource of the UN has been put to work; we've sifted the world's records to the very bottom, and not that day only but the time nine months earlier when these kids must have been conceived—only that doesn't fit either, because some of them were preemies as much as six weeks early, and they're the same, they're hollow, empty, drained . . . If we weren't at the end of our tether, I'd never have done such a crazy thing as coming to look for you. Because after all, I guess there isn't anything you can do to help, is there?"

The fire and rage which had burned in Chance when he arrived had died to ashes now, and he seemed to have no more words. Kotiwala stood thinking for a minute or more, and the villagers, growing restless, chattered among themselves.

At last the old man said, "The—the anti-senility drug. It's a success?"

"Oh yes. Thank God. If we hadn't some consolation in the midst of this mess I think we'd all go crazy. We've cut the death-rate fantastically, and because we planned well we can hope to feed the surplus mouths, and—"

"I think," Kotiwala broke in, "I can tell you what happened on the day we met."

Dazed, Chance stared at him. "Then for heaven's sake tell me! You're my last hope—you're *our* last hope."

"I can't offer hope, my friend." A sound like the echo of doom's own knell coloured the soft words. "But I can make what they call an educated guess. Did I not read once a calculation which showed that as many people are alive in this twenty-first century as have ever lived since the evolution of human beings?"

"Why—yes. I saw that myself, a long time back."

"Then I say that what happened on the day we met was this: the precise number of all the human beings there have ever been was exceeded for the very first time."

Chance shook his head in bewilderment. "I don't see—or—do I?"

"And it so turns out," Kotiwala continued, "that at the same time or very shortly afterward, you find, and make

available the world over, a drug which cures old age. Dr. Chance, you will not accept this, because I remember you made a kind of joke about an earthworm, but I do. I say that you have made me understand what I saw when I looked into the eyes of that new-born child, when I looked into the eyes of this little girl here." He touched the arm of the young mother at his side, and she gave him a shy quick smile.

"Not the lack of mind, as you have been saying. But the lack of a soul."

For a few seconds Chance imagined that he heard the hollow laughter of demons in the whisper of the winter wind. With a violent effort he rid himself of the delusion.

"No, that's absurd! You can't mean to make out that we've run short of human souls, as though they were stored up in some cosmic warehouse and issued off the shelf every time a child is born! Oh, come now, doctor—you're an educated man, and this is absurd."

"As you say," Kotiwala agreed politely. "That is something I won't venture to dispute with you. But I owe you my thanks, anyhow. You've shown me what I must do."

"That's great," Chance said. "Here I come half across the world hoping that you'll tell me what I must do, and instead you claim I've told you . . . What? What must you do?" A final flicker of hope leapt up in his face.

"I must die," said the sunnyasi, and took his staff, and his bowl, and without another word to anyone, even the young mother whom he had been comforting when Chance arrived, he set off with slow old-man's paces along the road that led to the tall blue mountains and the eternal ice by whose aid it was lawful for him to set free his soul.